

Kellen Cheng

✉ kellantan@alumni.princeton.edu ☎ 4074094268 🏠 Website 📧 kellantan 🇺🇸 U.S. Citizen

Education

Princeton University

Aug. 2022 - Dec. 2025

M.A. ELECTRICAL AND COMPUTER ENGINEERING, GPA: 4.000

RECEIVED MAY 2024

PHD QUALIFYING EXAM PASSED

PRINCETON, NJ

University of California, Los Angeles (UCLA)

Sept. 2018 - Jun. 2022

B.S. ELECTRICAL ENGINEERING, TECHNICAL BREADTH COMPUTER SCIENCE, GPA: 3.926

LOS ANGELES, CA

Experience

Microsoft: Member of Technical Staff

Oct. 2025 - Present

MANAGER: DR. MINGXI CHENG

MOUNTAIN VIEW, CA

- Designed an end-to-end synthetic data generation and post-training pipeline for aligning frontier multimodal LLMs on structured PowerPoint editing and reasoning tasks.
- Developed hybrid rule-based and LLM-driven perturbation strategies to generate 58k+ high-quality PowerPoint slide training pairs, enabling controlled evaluation of model reasoning and editing capabilities.
- Built data curation and quality filtering frameworks in collaboration with cross-functional teams, improving the fidelity and robustness of large-scale synthetic datasets.
- Developed multi-agent iterative refinement workflows to improve synthetic data quality across 20k+ PowerPoint deck creation samples spanning free-form and template-constrained generation.
- Ran large-scale RL-based post-training experiments and designed reward model graders to optimize model performance and alignment with human designer preferences across deck creation, deck editing, and slide editing scenarios.

IBM Research: AI Research Scientist Intern

May. 2025 - Aug. 2025

MENTOR: DR. ANNA LISA GENTILE

SAN JOSE, CA

- Proposed a method to generate type-constrained adversarial negative samples to improve LLM factuality and robustness.
- Built a scalable synthetic data pipeline leveraging structured source data to enable misinformation detection in long-context settings.
- Designed a GRPO-based self-play framework with a custom student-oracle reward function to model and inject realistic misinformation in long context training data.
- Led first-author research and filed two patents on multi-agent debate-based fine-tuning and adversarial factuality training.

Samsung Research America: NLP Research Scientist Intern

Jan. 2025 - May. 2025

MENTOR: DR. GANESH RAMESH

MOUNTAIN VIEW, CA

- Designed a multi-agent LLM framework for health dialogue summarization through iterative text refinement and feedback.
- Developed an unsupervised summarization approach that enabled compact language models to achieve competitive performance with Gemini-2.0-Flash across both open-source and proprietary datasets.
- Adapted the framework for on-device inference on Apple Silicon using Ollama and MLX-LM, requiring no GPUs and only 16 GB of system memory.
- Conducted parameter-efficient instruction fine-tuning with QLoRA across models ranging from 0.5B to 9B parameters.
- Published first-author research as an oral presentation at EACL 2026 Main.

IBM Research: AI Research Scientist Intern

Jun. 2024 - Sept. 2024

MENTOR: DR. ANNA LISA GENTILE

SAN JOSE, CA

- Synthesized and curated an evaluation benchmark for health advice guardrails from Common Crawl web data.
- Designed a sparse human-in-the-loop framework for scalable, semi-automatic annotation of synthetic data.
- Formulated a method to automatically generate synthetic data using compact LLMs for health advice guardrails.
- Fine-tuned compact detector models on a mixture of synthetic and open-source training data, outperforming GPT-4o by 3.73% in accuracy and 1.54% in F1 with 400x fewer parameters.

- Published first-author research at EMNLP 2024 Industry Track and IJCNLP-AAACL 2025 Main, and filed a patent on continual learning and knowledge distillation for AI safety guardrails.

Princeton: Graduate Researcher

Nov. 2022 - Dec. 2025

ADVISOR: DR. SUMA BHAT

PRINCETON, NJ

- Developed an end-to-end two-mask infilling fine-tuning objective for injecting idiomatic knowledge into pre-trained language models using the IEKG dataset.
- Developed a two-stage transfer learning approach that achieved 83.75% accuracy on the IMPLI benchmark, improving upon the previous state of the art by 12%.
- Conducted ablation and data perturbation studies across language models ranging from 0.5B to 7B parameters, revealing that several off-the-shelf models performed better when contextual information was removed.
- Published first-author research at NAACL 2024 Main and co-authored research at EMNLP 2023 Main.

Selected Publications

- Kellen Tan Cheng**, Ganesh Ramesh, Nafiul Rashid, Geoffrey Jay Tso, Jilong Kuang. *Compact Language Models with Iterative Text Refinement for Health Dialogue Summarization*. EACL 2026 Main (Oral).
- Kellen Tan Cheng**, Anna Lisa Gentile, Chad DeLuca, Guang-Jie Ren. *STAR: Self-Automated Back-Querying for Production Data Generation*. IJCNLP-AAACL 2025 Main.
- Chad DeLuca, Anna Lisa Gentile, Shubhi Asthana, Bing Zhang, Pawan Chowdhary, **Kellen Tan Cheng**, Basel Shbita, Pengyuan Li, Guang-Jie Ren, Sandeep Gopisetty. *OneShield - the Next Generation of LLM Guardrails*. ArXiv 2025.
- Kellen Tan Cheng**, Anna Lisa Gentile, Pengyuan Li, Chad DeLuca, Guang-Jie Ren. *Don't Be My Doctor! Recognizing Healthcare Advice in Large Language Models*. EMNLP 2024 Industry Track.
- Kellen Tan Cheng**, Suma Bhat. *No Context Needed: Contextual Quandary In Idiomatic Reasoning With Pre-Trained Language Models*. NAACL 2024 Main.
- Ziheng Zeng, **Kellen Tan Cheng**, Srihari Venkat Nanniyur, Jianing Zhou, Suma Bhat. *IEKG: A Commonsense Knowledge Graph for Idiomatic Expressions*. EMNLP 2023 Main.

Awards & Organizations

Toby & Jack Wolf Travel Grant	2024
Bede Liu Travel Grant	2023
Princeton ECE Departmental Fellowship	2022
Tau Beta Pi	2020 - Present
IEEE Eta Kappa Nu (HKN)	2019 - Present
UCLA Dean's Honor List	2019 - 2022

Service

EACL 2026 Session Chair	2026
ICML Reviewer	2026
NeurIPS Reviewer	2026

Skills

Languages	Python, C++, MATLAB
Frameworks	PyTorch, Tensorflow, MLX, Transformers, TRL
Tools	Ollama, Slurm, LSF, Azure, AWS EC2, Conda, Jupyter, LaTeX, Git, WSL, MS Office, GitHub Copilot CLI